

PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience building secure and scalable web applications using React.js, Node.js, Express.js, MongoDB, and MySQL. Currently enrolled in the Digilians Software Engineering Program, specializing in Full-Stack Development, AWS Cloud Computing, DevOps, and Secure Web Development.

Experienced in designing RESTful APIs, implementing JWT-based authentication and authorization systems, integrating cloud services, and applying OOP, SOLID, Design Patterns, Data Structures, and Algorithms. Passionate about building reliable, secure, and cloud-ready applications while continuously expanding expertise in DevOps and Web Security.

TECHNICAL SKILLS

- **Languages:** JavaScript, C++, MySQL, python
- **Frontend:** React.js, Redux Toolkit, HTML5, CSS3, Bootstrap, React Router
- **Backend:** Node.js, Express.js, RESTful APIs, JWT Authentication, Refresh Token Rotation, Cookie-Based Authentication, MVC Architecture
- **Databases:** MongoDB, Mongoose, MySQL
- **Cloud & DevOps:** AWS Practitioner (EC2, S3, IAM, RDS, EBS, CloudFront), aws developer(python based), aws SysOps, Docker, Linux, Git, GitHub, CI/CD, Docker file, Terraform
- **Security:** OWASP Top 10, XSS Prevention, CSRF Protection, Secure Authentication & Authorization, Input Validation & Sanitization
- **Software Engineering:** OOP, Data Structures, Algorithms, SOLID Principles, Design Patterns, System Design Fundamentals
- **Tools:** Postman, Axios, Joi, Swagger, Socket.io, VS Code, workbench, VMware, MongoDB Compass

EXPERIENCE

Digilians Software Engineering Trainee

Dec 2025 – Present

MCI – Cairo, Egypt

- Building full-stack applications using React.js, Node.js, Express.js, MongoDB, and MySQL.
- Developing cloud-based solutions using AWS services and modern deployment practices.
- Implementing secure authentication systems using JWT, Refresh Tokens, and role-based authorization.
- Applying software engineering best practices, design patterns, and clean architecture principles.
- Studying DevOps fundamentals including Docker, Linux, CI/CD workflows, and cloud operations.
- Practicing secure coding techniques and web application security principles based on OWASP standards.

PROJECTS

S-WINDs – Smart Weather-Integrated Navigation & Driving System *React.js, Redux Toolkit, Node.js, Express.js, MongoDB, JWT, Socket.io, PWA*

- Engineered a full-stack PWA combining real-time route planning with ETA-based hyper-local weather forecasting, improving driving safety through waypoint-level weather risk scoring.
- Implemented production-grade JWT authentication with Access/Refresh Token Rotation, device fingerprinting (UA-Parser), and GeoIP country-change detection to prevent session hijacking.
- Built a Geohash-based weather caching layer using MongoDB TTL indexes (30-min expiry), reducing Tomorrow.io API calls by an estimated 60–70% under concurrent load.
- Integrated OSRM routing engine to compute per-waypoint ETAs and built a custom risk engine (calculateSafeSpeed, calculateRiskLevel) adapting recommendations per vehicle type.
- Integrated Socket.io for real-time fleet tracking and live emergency alert broadcasting.

Bug Bounty Platform

React.js, Node.js, Express.js, MongoDB, JWT

- Developed a role-based vulnerability disclosure platform (hunter, company, admin) with JWT cookie authentication and role-gated API routes implementing RBAC design patterns.
- Designed a dynamic reputation engine auto-incrementing hunter scores on report acceptance with severity-weighted allocation (critical: 100pts, high: 50pts, medium: 25pts, low: 10pts).
- Structured backend across 6 domain controllers and 5 Mongoose models with ownership checks preventing unauthorized access, implementing OWASP A01 access control principles.
- Applied Joi input validation across all endpoints and centralized error handling middleware ensuring consistent, secure API responses.

Super Note – Intelligent Note Management System

React.js, Node.js, Express.js, MongoDB, JWT, Node-Cron, MVC

- Built a full-stack note management system featuring Soft Delete with Trash Recovery, per-note Version History, and automated Cron Job-based expiry cleanup executing every hour.
- Implemented JWT authentication, bcrypt password hashing, and cookie-based sessions with Joi validation across all protected RESTful API endpoints following MVC architecture.
- Built an NLP keyword extraction engine with a bilingual stopwords list (Arabic + English) for auto-tagging, and a Crypto-based share token system for secure note sharing.
- Engineered a Higher-Order Function activity tracker (track()) wrapping controllers to auto-log all user actions to MongoDB without modifying individual controller logic.
- Optimized query performance using Compound Indexes on (isDeleted + expiresAt), reducing scheduled cleanup scan time from O(n) to O(log n) at scale.

PDF Pro License Server

Node.js, Express.js, JWT, node-machine-id

- Developed a software licensing platform using hardware fingerprinting (node-machine-id) to enforce machine-bound, single-device license activation.
- Built backend APIs for cryptographic license key generation, verification, and full lifecycle management (block/unblock/reset) with admin-protected endpoints.
- Designed an integrated Admin Dashboard for real-time license status monitoring and applied one-time activation workflows with tamper-resistant verification logic.

EDUCATION

Bachelor of Arts – Psychology

Zagazig University, Egypt

2018

LANGUAGES

Arabic: Native **English:** C1 – Professional Working Proficiency.